GRS-Det: An Anchor-Free Rotation Ship Detector Based on Gaussian-Mask in Remote Sensing Images

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***Abstract*— Ship detection is a significant and challenging task in remote sensing. Due to the arbitrary-oriented property and large aspect ratio of ships, most of the existing detectors adopt rotation boxes to represent ships. However, manual-designed rotation anchors are needed in these detectors, which causes multiplied computational cost and inaccurate box regression. To address the abovementioned problems, an anchor-free rotation ship detector, named GRS-Det, is proposed, which mainly consists of a feature extraction network with selective concatenation module (SCM), a rotation Gaussian-Mask model, and a fully convolutional network-based detection module. First, a U-shape network with SCM is used to extract multiscale feature maps. With the help of SCM, the channel unbalance problem between different-level features in feature fusion is solved. Then, a rotation Gaussian-Mask is designed to model the ship based on its geometry characteristics, which aims at solving the mislabeling problem of rotation bounding boxes. Meanwhile, the Gaussian-Mask leverages context information to strengthen the perception of ships. Finally, multiscale feature maps are fed to the detection module for classification and regression of each pixel. Our proposed method, evaluated on ship detection benchmarks, including HRSC2016 and DOTA Ship data sets, achieves state-of-the-art results.**

***Index Terms*— Anchor-free method, remote sensing images (RSIs), rotation object detection, ship detection.**

I. Introduction

With the rapid development of optical remote sensing (RS) technology and computer vision, the research of optical RS images (RSIs) has been attracting a lot of attention. Among them, ship detection is an important subject in the field of RS for its military, commercial, and civilian value. Many methods for ship detection have been proposed; however, it is still a challenging task due to the complexity of the RS scenes and the large differences of the ship scale.

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In traditional methods, ship detection is divided into three stages. First, the sea–land segmentation is performed to eliminate the interference of the land part [1]–[5]. Then, candidate regions are generated to locate areas that may contain ships [4]–[6]. Finally, feature extraction algorithms [7]–[10] are used to extract and classify candidate regions to identify the ship. Although traditional methods have achieved promising results, their performances are poor in complex environments.

Influenced by the excellent performance of the deep convolutional neural network (CNN) [11] on image classification [12], a series of general object detection algorithms [13]–[23] based on CNN are presented, which are usually divided into two streams: two- and single-stage methods. Two-stage methods [13]–[17] focus on the precision of detectors that contain two parts: region proposals generation and region proposals' classification. On the contrary, one-stage methods [18]–[23] pay more attention to the speed of detectors.

Most of these general detectors have been widely employed for RS object detection. However, these detectors based on horizontal bounding boxes (HBBs) are limited to represent objects with large aspect ratios, especially for ships. Besides, unlike natural images, the ships in RS are usually dense distribution, as shown in Fig. 1. The two characteristics result in false detection and redundancy representation when using HBBs. To solve these problems, a lot of oriented detectors [24]–[33] are designed, which have shown remarkable success. However, these methods rely on predefined anchors whose sizes, aspect ratios, and rotation angles are needed to be designed in the preprocessing phase, which results in numerous hyperparameters. Moreover, these hyperparameters are closely related to data set and largely affect detection performance, which makes detectors inflexible and vulnerable.

To overcome the abovementioned problems and make full use of the characteristics of ships, an anchor-free rotation ship detector called GRS-Det is proposed. Our detector transforms the ship detection task into a binary semantic segmentation task to detect dense ships, which is composed of a U-shape feature extraction network with a selective concatenation module (SCM), a Gaussian-Mask model, and a fully convolutional network (FCN)-based detection module. Benefiting from utilizing the anchor-free method, GRS-Det is more efficient and flexible than existing methods.

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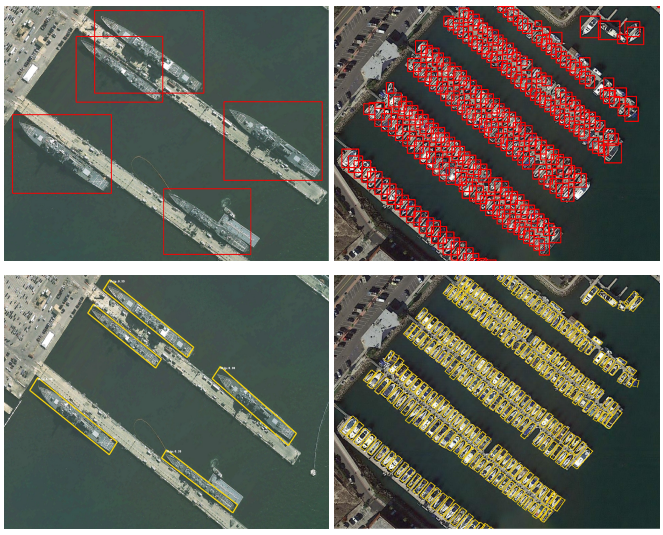

Fig. 1. Visualization of ships in different scenes. Large in aspect ratio and dense in distribution, the HBB is not suitable to represent ships, as shown in the first row.

First, in order to tackle the problem of large differences of the ship scale in RS, a U-shape network is constructed for multiscale feature fusion. In the feature fusion stage, due to the divergence of different layer channels, a channel unbalance problem between high- and low-level feature maps is generally faced. In order to solve this problem, a simple but efficient module SCM is designed to merge different-layer features. The final output of the feature extraction network is fed to the detection module.

Second, considering the geometry characteristics of ships, a rotation Gaussian-Mask is proposed to model ship. Although using a rotation bounding box to represent ship largely avoids background information, many background areas still exist in the head and tail of ships. Therefore, a rotation Gaussian-Mask is presented to model the ships accurately, and the aspect ratio of ships is adopted to control the covariance of confidence distribution. The rotation Gaussian-Mask ensures that the ship center has the highest reliability, and the confidences of other regions are gradually reduced by the Gaussian distribution. Similar to the soft-label method in [34], the Gaussian-Mask solves the problem of background interference when using a binary map to represent ships. Meanwhile, using the gradually changed confidences, the model can learn the implied distance information between each location and the center of the ship.

Third, inspired by previous work [35], [36], the ship detection task is transformed into a binary semantic segmentation task to directly detect the pixels belonging to ships and predict the distance between each pixel and four boundaries. The output of the feature extraction network is fed into FCN-based detection module to predict the pixel-level kernel map, region map, angle map, and geometry maps that are assembled into detection results. The kernel map and the region map are responsible to predict the confidence of the kernel area and the whole area of ships, respectively. The angle and geometry maps are used to compose detection bounding boxes. In the training phase, the binary map and Gaussian-Mask are adopted to supervise the kernel and region maps, respectively, that utilize the context information of ships to further improve performance. Finally, our GRS-Det is evaluated on two public ship detection data sets (i.e., HRSC2016 [37] and DOTA Ship [38]). Compared with the state-of-the-art approach, our method achieves better performance.

The contributions can be summarized as follows.

1) An anchor-free rotation ship detector named GRS-Det is proposed. By converting ship detection into a binary semantic segmentation task, it can directly detect dense rotation ships.
2) For the issue of a large difference in the ship scale, a U-shape network with SCM is adopted to fuse multilevel features. With the help of SCM, the channel unbalance problem in feature fusion is solved, which further improves performance.
3) A rotation Gaussian-Mask model is designed to deal with the mislabeling problem of rotation bounding boxes. The aspect ratio of the ship is used to control the covariance of confidence distribution.

The remainder of this article is organized as follows. In Section II, we briefly review related works on ship detection methods, oriented detection methods, and anchor-free methods. In Section III, we describe the details of our proposed detection framework GRS-Det. We introduce the data set and evaluation metrics in Section IV and show the experiment settings and results in Section V. Finally, Section VI concludes this research in this article.

II. Related Work

Ship detection methods in RSIs can be divided into traditional and CNN-based methods. Traditional methods consist of three stages: sea–land segmentation [1]–[5], region proposals generation [4]–[6], and region classification [7]–[10]. Tang *et al.* [4] adopted a compressed domain for fast region proposals generation and utilized deep neural network (DNN) to perform feature representation and classification on each region proposal. Liu *et al.* [6] constructed a rotated ship space and used a two-cascaded linear model to capture ships. Yang *et al.* [9] utilized pixel and region statistic characteristics to select ship candidates. These methods are widely employed for their promising results. However, each stage of them needs handcraft, which is inefficient.

With the advance of CNN techniques, a host of CNN-based object detection methods has been proposed, which are usually divided into two streams: horizontal and oriented detection methods. Horizontal detection methods [13]–[23] are widely used in natural images. Among them, the most commonly used detector is Faster R-CNN [15], in which a region proposal network (RPN) is utilized to generate candidate proposals and a region-of-interest (RoI) pooling layer [14] is adopted to extract each proposal region. With the help of the RPN, Faster R-CNN [15] achieves end-to-end detection and improves speed and precision. In order to satisfy real-time demand in practice, you only look once (YOLO) [18]–[20] and single-shot multibox detector (SSD) [21]–[23]

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were proposed. Following the thought of regression, YOLO [18]–[20] predicts the category and location of objects in each grid of image directly, without any region proposal. Inspired by the thought of RPN and multiscale, SSD [21]–[23] effectively combines these ideas to achieve faster speed and comparable precision to [15]. Benefiting from the powerful feature representation ability of CNN, these methods greatly outperform traditional methods in RS ship detection.

However, unlike natural images, the ships in RSIs are usually dense distribution and arbitrary oriented, as shown in Fig. 1. Therefore, it is prone to result in false detection and redundancy representation with horizontal detection methods. To overcome these problems, some oriented detection methods have been proposed [24]–[33]. In rotated region-based CNN (RR-CNN) [24], a rotated RoI pooling layer is used to extract the feature of the rotation object, which reduces a lot of redundancy areas. In order to enhance the quality of the proposal, rotation RPNs (RRPNs) [25] improve RPN in Faster RCNN by using rotation anchors. Rotation dense feature pyramid networks (R-DFPNs) [28] enhance the detection capability by connecting the multiscale feature maps densely. However, a large number of the computational cost is introduced to the feature extraction network due to dense connection. The image cascade network (ICN) [29] achieves excellent precision by using both the image cascading and feature cascading. Ding *et al*. [31] proposed an RoI-Transformer to learn the parameters of rotation so that the network can accurately acquire the features of the rotation object while maintaining a fast speed. Based on RetinaNet [39], the refined rotation RetinaNet (R3Det) [32] adds the feature refine module and rotation module that ensures both speed and accuracy simultaneously. It achieves state-of-the-art results on multiple data sets. Although these methods have shown remarkable success, they cannot avoid the manual design of anchor, which is inefficient and inflexible. Especially, there are two difficulties in existing oriented detectors as follows.

1) Since sampling objects with rotation anchors is required in existing methods, the extra parameter angle is introduced in the anchor design process, which causes the number of anchors to be multiplied. It tremendously increases computational complexity and memory consumption.
2) Since the objects in RSIs are arbitrary oriented, it is difficult for high-quality sampling with manual-designed rotation anchors, which results in inaccurate regression.

Recently, a host of anchor-free methods booms, in general, object detection and scene text detection [35], [36], [40]–[42]. These methods are usually divided into two categories: keypoint-based and segmentation-based methods. For keypoint-based methods, Law and Deng [40] proposed CornerNet to locate objects through predicting the upper left and lower right corners of the bounding box. Based on CornerNet [40], Zhou *et al*. [41] proposed ExtremeNet that locates objects by predicting the top, bottom, left, and right extreme points and achieves improving performance. For segmentation-based methods, DenseBox [35], the first anchor free method, uses FCN to predict bounding box and confidence straightly. However, the image pyramid strategy is used in DenseBox to detect different scales' objects, which is greatly against the philosophy of FCN. Following the idea of [35], FCN structure is also used in [36] and [42] to predict objects densely. Efficient and accuracy scene text detector (EAST) [36] regresses the distance between each pixel inside objects and borders, which can handle oriented texts. Fully convolutional one-stage object detector (FCOS) [42] adopts the same backbone as RetinaNet [39] and proposes center-ness to suppress the low-quality detected bounding boxes. However, they are not suitable for RS due to the dense distribution and complex scenes. There are only a few studies for anchor-free methods in RS. Interacting embranchment network (IENet) [43] modifies the FCOS [42] structure with an oriented regression branch to detect oriented objects. The oriented objects detection network (O^2-DNet) [44] adopts a pair of middle lines to represent objects. None of them considers the characteristics of ships. The purpose of our work is to design an anchor-free oriented detector for ships.

III. Our Method

In this section, we will introduce our GRS-Det in detail. Our detection framework is shown in Fig. 2, which contains a U-shape feature extraction backbone with SCM, a Gaussian-Mask ship model, and an FCN-based detection module.

A. Feature Extraction With SCM

Generally, CNN needs multiple downsampling to get high-level feature maps for the detection task, which has large receptive fields and strong semantics. However, the resolution is so low that small objects are lost. In contrast, low-level feature maps have high resolution and rich geometric details that are useful for small objects. Therefore, it is necessary to fuse multilevel features.

As shown in Fig. 2, ResNet [45] is used to extract the feature of the input image, and a series of feature maps $\{C_1, C_2, C_3, C_4\}$ are obtained from each stage whose number of channels are {64, 256, 512, 2048}, respectively. After that, a top–down strategy is used to fuse features. The minimum-size feature map C_4 is upsampled by bilinear interpolation and fused with the feature map C_3, and then, the merged feature F_1 is obtained. Repeating these operations, we will obtain the fused feature maps $\{F_1, F_2, F_3\}$ whose numbers of channels are {128, 64, 32}, respectively.

In the fusion step, due to the divergence of channel numbers, an unbalance problem between the high- and low-level features is faced. For example, in the first fusion step, the channel numbers of high- and low-level features are 2048 and 512, respectively, which results in a channel unbalance problem. Inspired by [46], an SCM is proposed to merge different-level features fairly. In SCM, a channel attention method that adds weights on each channel is adopted. The structure of the SCM is presented in Fig. 2. The high-level feature map is upsampled by bilinear interpolation and concatenated with a low-level feature map directly. Then, a global average pooling (GAP) is used to encode the concatenated feature into a vector, and this vector is fed into a fully connected

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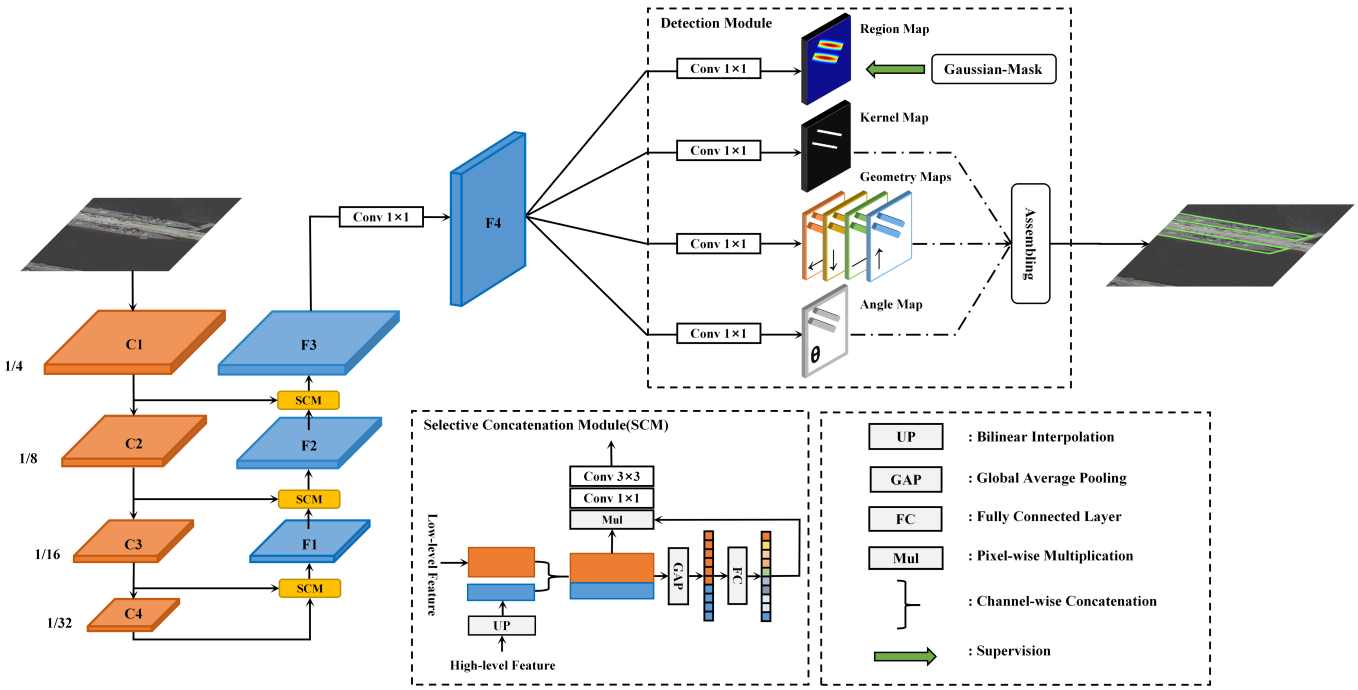

Fig. 2. Overview of GRS-Det. It consists of a U-shape feature extraction network with SCM, a rotation Gaussian-Mask model, and an FCN-based detection module. The rotation Gaussian-Mask is proposed to model ship much properly (detail is given in Section III-B), and the detection module is described, in detail, in Section III-C.

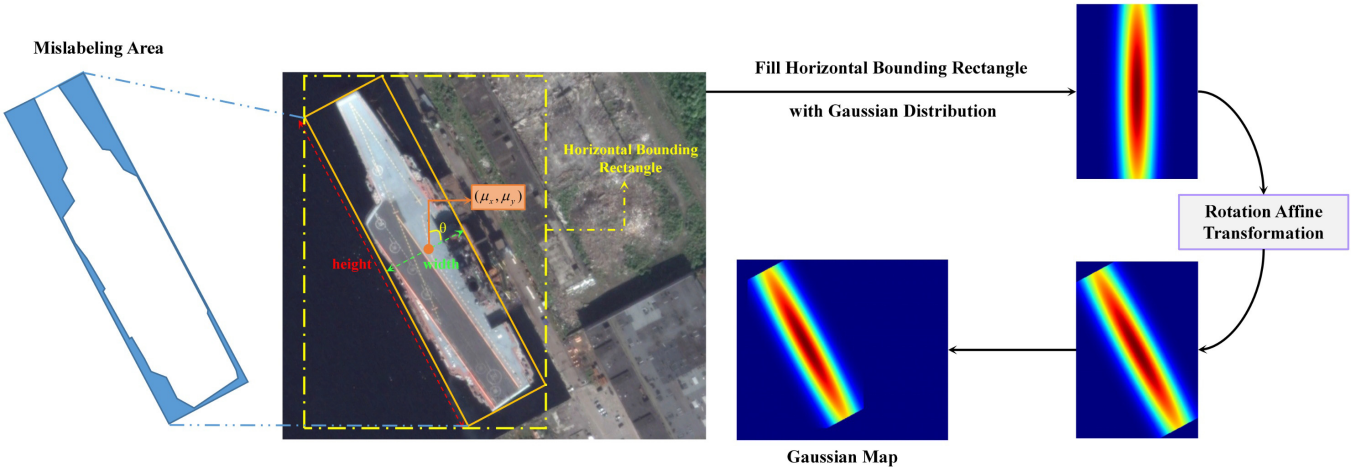

Fig. 3. Detailed process of the rotation Gaussian-Mask ship model. Take the horizontal bounding rectangle of ship and fill each pixel following the law of Gaussian distribution whose covariance is controlled by the aspect ratio of ship. Finally, making a rotation affine transform to attain the Gaussian map.

layer with sigmoid activation. Next, this vector is multiplied with the abovementioned concatenated feature map, and a 1×1 convolutional operation is performed to eliminate the semantic gap. Finally, after a 3×3 convolutional operation, we can get the balanced feature map F_i.

B. Rotation Gaussian-Mask Model

Due to the large aspect ratio and dense distribution of ships, representing ships with HBBs usually contains a lot of background information and the redundancy region. As shown in Fig. 1, for densely arranged ships, using HBB results in a large Intersection over Union (IoU) between different objects. For large ships, nearly half part inside the bounding box is the background, which will extremely affect the identification of ships. Therefore, the rotation bounding box is adopted to alleviate the interference. However, the problem of including the background is not completely resolved by using a rotation bounding box, as shown in Fig. 3. An inevitable mislabeling problem is introduced that many background areas are usually contained in the tail and head of the ship because the shape of ships is similar to a shuttle (the tail and head are narrow, and the body is wide).

To alleviate the mislabeling problem, according to the ship's geometric characteristics, a rotation Gaussian-Mask is presented to model the ship. The aspect ratio of the ship is used to control the covariance of confidence distribution.

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Meanwhile, our proposed rotation Gaussian-Mask ensures that the ship center has the highest confidence, and the confidences of remaining regions are gradually reduced in accordance with the law of Gaussian distribution. Different from using a binary map to represent the ship, this rotation Gaussian-Mask not only solves the mislabeling problem but also enables the model to capture the location information.

During the modeling process, the center of the rotation bounding box is assigned as the ship center with the highest confidence, and then, each pixel is filled following the law of Gaussian distribution. In detail, as illustrated in Fig. 3, we first take the horizontal bounding rectangle of the ship and define ship's long side and short side as the height and width, respectively. Then, with the Gaussian distribution, the confidence of each pixel is generated according to the formula as follows:

$$G(x, y) = \mathrm{Exp}\left(-\left(\lambda_w \cdot \frac{(x-\mu_x)^2}{2\sigma_w^2} + \lambda_h \cdot \frac{(y-\mu_y)^2}{2\sigma_h^2}\right)\right) \quad (1)$$

where μ_x and μ_y denote the coordinates of ship center, σ_h and σ_w are proportional to the height and width of the ship, respectively, as in [47], and λ_h and λ_w represent the covariance control factors determined by the aspect ratio (height to width), given by

$$\begin{cases} \lambda_h(\mathrm{ratio}) = \alpha \cdot \mathrm{ratio}^{-1} \\ \lambda_w(\mathrm{ratio}) = \beta \cdot \mathrm{ratio} \cdot e^{-\xi \cdot \mathrm{ratio}}. \end{cases} \quad (2)$$

In this article, we set $\alpha = 0.1$, $\beta = 2.95$, and $\xi = 0.35$ by experimental experience in order to build a more accurate ship model. Finally, according to the rotation angle of the ship, an affine transformation is carried out between the horizontal Gaussian-Mask and the rotation Gaussian-Mask. Benefited from the Gaussian-Mask model, the location information is introduced to the detector, which alleviates the mislabeling problem and further improved the performance.

C. Detection Module

As described in Fig. 4, the multiscale features obtained by the feature extraction network are sent to the detection module, and then, a series of prediction maps are generated. The maps consist of the kernel map, region map, angle map, and geometry maps. The kernel and region maps are responsible for predicting the confidence of ships, but different types of labels are applied to guide them. The angle and geometry maps are used to generate detection bounding boxes at the location where the confidences of the pixels in the kernel map are greater than the confidence threshold.

1) Training Phase: Since GRS-Det is a pixel-based detector and the precise pixel-level labels consume a lot of manpower, it is necessary to make full use of the ship rotation label in different ways.

For the kernel map, we first generate a shrunk box by shrinking two short sides of each rotation box and then fill them with value 1. Formally, given the four sides of the rotation rectangle (S_{long1}, S_{short1}, S_{long2}, and S_{short2}), the shrunk box can be obtained by (S_{long1}, $(1-\gamma_s)\cdot S_{\mathrm{short1}}$, S_{long2}, $(1-\gamma_s)\cdot S_{\mathrm{short2}}$), and γ_s is set to 0.3 in our experiments (ablation study in Section V illustrates why 0.3 is chosen). With this generation strategy, it can be ensured that only the high confidence areas are applied to regress the bounding boxes.

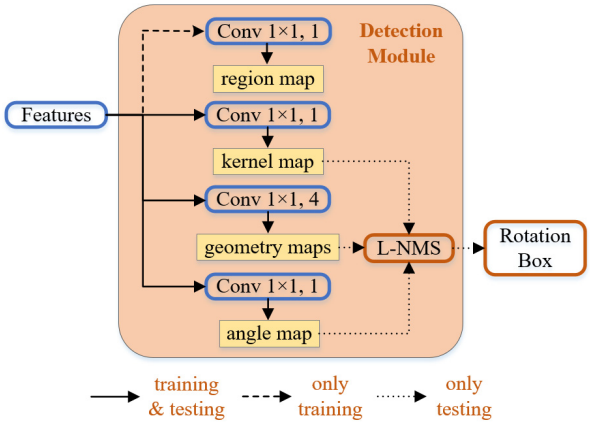

Fig. 4. Illustration of detection module. This module combines the kernel map, angle map, and four geometry maps to predictions and gets the detection results after L-NMS.

Due to the shrink strategy in the kernel map, the boundary areas that generally contain much context information cannot be utilized. Therefore, the region map is added to provide context information that the kernel map lacks. Because of the mislabeling problem in the peripheral of ships, the above-proposed Gaussian-Mask is adopted as ground truths to alleviate the mislabeling problem and capture location information. Different from the kernel map, the region map is only used to provide supervision information during the training stage. With the help of the two maps, the rotation label is fully utilized, which helps the detector achieve more precise results.

The general rotation object detection method samples the ground truths by manually designing anchors, and the performance of the detector is sensitive to the quality of anchors, which results in many hyperparameters and huge time costs. To overcome these problems, GRS-Det transforms rotation bounding box into geometry maps and angle maps. As shown in Fig. 5, we first shrink the rotation bounding boxes, such as the kernel map. Then, for each positive pixel, we calculate its vertical distances to the four boundaries of the rotation bounding box, respectively, and use them to fill geometry maps. Similarly, for the angle map, we fill each positive pixel with the rotation angle of the ship.

2) Inference Phase: Formally, given a kernel map $K \in (0, 1)^{H\times W}$, we extract all the points with a value greater than the threshold τ. These points are considered belonging to ships and are represented as $\mathcal{K}$. According to the points in $\mathcal{K}$, we extract the corresponding points of geometry maps and angle map, and the collections of them are called $\{G^{(T)}(\mathcal{K}), G^{(R)}(\mathcal{K}), G^{(B)}(\mathcal{K}), G^{(L)}(\mathcal{K}), G^{(\theta)}(\mathcal{K})\}$, which represents the distance between the top, right, bottom, and left sides and the rotation angle of ships, respectively.

Assembling these predicted values, we obtain many candidate rotation bounding boxes. In order to take full advantage of predictions and enhance the robustness of the detector,

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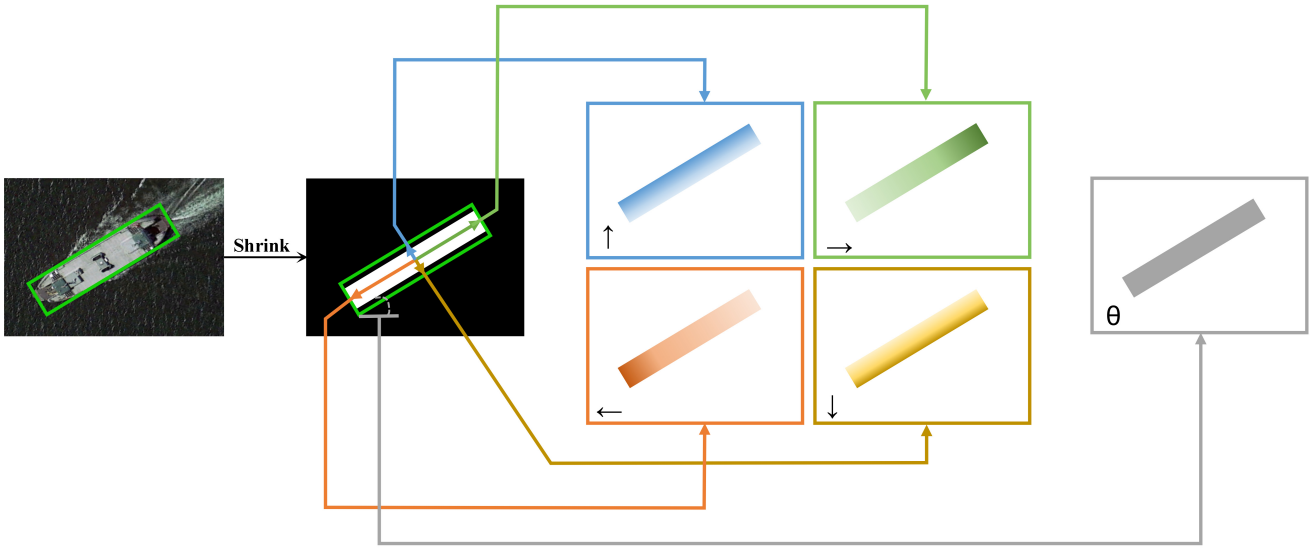

Fig. 5. Illustration of the training label generation process of geometry maps and angle map. Four color maps represent the distance between each pixel inside kernel area and different boundary, respectively. The gray map represents the rotation angle of the ship.

local-aware nonmaximum suppression (L-NMS) [36] is followed to obtain the final detection results. With an idea similar to voting, this method alleviates the effect of prediction errors of individual points and is much suitable for dense predictions. The overall process is summarized in Algorithm. 1.

Algorithm 1 Detection Module

Input: Kernel and Geometry maps of an image: K, $G^{(T)}$, $G^{(R)}$, $G^{(B)}$, $G^{(L)}$, $G^{(\theta)}$; Kernel map and IoU threshold: τ, ε
Output: The set of rotation bounding box with score: S

```
// Extract points greater than threshold
K ⇐ ExtractPoints(K, τ)
S' ← ∅
for each k ∈ K do
  t ← G^(T)(k)
  r ← G^(R)(k)
  b ← G^(B)(k)
  l ← G^(L)(k)
  θ ← G^(θ)(k)
  score ← value(k)
  // Add rotation bounding box with score
  S' ← S' ∪ {Assemble(t, r, b, l, θ), score}
end for
// Local-aware Non-Maximum Suppression
S ← ∅
for each s1, s2 ∈ S' do
  // Merge
  if IoU(s1, s2) > ε then
    s = (score_s1 * s1 + score_s2 * s2)/(score_s1 + score_s2)
    S ← S ∪ {s}
  end if
end for
// Standard Non-Maximum Suppression
S ← NMS(S)
return S
```

D. Loss Function

Since our GRS-Det is an end-to-end detector, the multitask loss function is used, which is defined as follows:

$$L = L_{\text{Reg}} + \lambda L_{\text{Cls}} \tag{3}$$

where L_{Reg} and L_{Cls} represent the regression and classification loss of GRS-Det, respectively, and the hyperparameter λ is used to balance the importance of the two terms. In our experiment, λ is set to 0.01.

1) Classification Loss: The classification loss consists of two parts, the loss of the kernel map and the region map, named L_{kernel} and L_{region}. λ_k and λ_r are used to control the ratio of the two losses. In the experiment, we set them to 1 and 0.5, respectively

$$L_{\text{Cls}} = \lambda_k L_{\text{kernel}} + \lambda_r L_{\text{region}}. \tag{4}$$

Since objects in RSIs are normally small, L_{kernel} adopts the dice loss [48] to relieve the foreground and background unbalance problem. It is expressed as follows:

$$L_{\text{kernel}} = 1 - \frac{2\sum_i K_{\text{pred}}(i) * K_{\text{gt}}(i)}{\sum_i K_{\text{pred}}(i) + \sum_i K_{\text{gt}}(i)} \tag{5}$$

$$L_{\text{region}} = \sum_i \text{SmoothL1}\big(R_{\text{pred}}(i) - R_{\text{gt}}(i)\big) \tag{6}$$

$$\text{SmoothL1}(x) = \begin{cases} 0.5\ x^2, & |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{7}$$

where K_{pred} denotes the predictions of the kernel map and K_{gt} denotes the ground truths of the kernel map. For the region map, SmoothL1 is performed as the classification loss, where R_{pred} and R_{gt} denote the prediction and ground truth of the region map, respectively.

2) Regression Loss: The whole regression loss consists of the bounding box regression loss and angle regression loss. Due to large differences in ship scale in RSIs, IoU loss is

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selected and combined with a simple angle regression loss as the final regression loss. The loss function is given as follows:

$$L_{\text{Reg}} = -\log \frac{|\text{Area}_{\text{pred}} \cap \text{Area}_{\text{gt}}|}{|\text{Area}_{\text{pred}} \cup \text{Area}_{\text{gt}}|} + \lambda_\theta \left(1 - \cos\left(\theta_{\text{pred}} - \theta_{\text{gt}}\right)\right) \quad (8)$$

where $\text{Area}_{\text{pred}}$ is the area of the predictions, Area_{gt} is the ground truths, θ_{pred} is the predicted angle, θ_{gt} is the true rotation angle of ships, and the tradeoff coefficient λ_θ is set to 20.

IV. Data Sets and Evaluation Metrics

In this section, we will introduce the data sets and evaluation metrics used in our experiments. To validate the performance of the proposed method, we evaluate our proposal on two public ship data sets: HRSC2016 [37] and DOTA Ship [38].

A. HRSC2016

HRSC2016 [37] is a challenging ship data set that contains images from two scenarios: ships on the sea and ships close inshore. All the images are collected from six famous ports. In the data set, the image resolution is between 2 and 0.4 m, the image size is between 300 × 300 and 1500 × 900, and most of them are larger than 1000×600. The training, validation, and test data sets consist of 436, 181, and 444 images, respectively.

B. DOTA Ship

DOTA [38] is a large-scale data set for object detection in RSIs. It contains 2806 images from different sensors and platforms. Each image is about 800×800–4000×4000 pixels in size and contains objects with a wide variety of scales, orientations, and shapes. These DOTA images are annotated by experts in aerial image interpretation using 15 common object categories. We choose 434 annotated images that contain 37 028 ship objects and divide these images into training data set and test data set randomly, which includes 326 and 108 images, respectively. Since the original image of DOTA [38] is too large, we split it into 1024×1024. After splitting, the training data set includes 1727 images, the testing data set includes 575 images, and 71 202 ships are contained in total. The details are given in Table I.

TABLE I
Detail of the DOTA Ship Data Set

DOTA Ship	Number of Images	Number of Ships
Training Set	1727	52827
Test Set	575	18375
Total	2302	71202
Distribution of Size		
10-50 pixel	50-300 pixel	≥300 pixel
57%	41%	2%

C. Evaluation Metrics

In this article, the widely used measures in object detection, precision, recall, F1-measure, and average precision (AP) are adopted to evaluate the performance of the ship detector. Moreover, we use the number of parameters (Params) and average running time to evaluate the complexity and speed of the model, respectively.

V. Experiments

A. Implementation Details and Parameter Optimization

In this article, all the experiments are implemented under the Tensorflow framework. For both the HRSC2016 [37] and DOTA Ship data sets [38], the input image is resized to 800×800, and the pixels are padded with zeros to keep the aspect ratio. Then, the network is trained on single NVIDIA GeForce GTX 1080Ti with 12-GB GPU memory, and the batch size is set to 8.

For the network initialization, the weights pretrained on ImageNet are used to initialize the backbone (i.e., ResNet), and other added layers are randomly initialized by a zero-mean normal distribution with a standard deviation of 0.01. Adam [49] is selected as the optimizer for our model, and the learning rate is set to 0.0001 at the beginning. For every 20k iterations, the learning rate is multiplied by 0.94, and the model is trained by 80k iterations in total. In the training phase, we use simple data augmentations, including random horizontal flipping, random crop, and random scale, to enhance the robustness of the detector. In the inference phase, the threshold of the kernel map is set to 0.1, which ensures as large recall as possible.

B. Ablation Study

We conduct comprehensive experiments to evaluate the contribution of each component in GRS-Det. GRS-Det without SCM and Gaussian-Mask is selected as the baseline, and the same training and data augmentation strategies are adopted in all experiments. All the ablation experiments are implemented on HRSC2016 [37].

The module, SCM, is proposed to tackle the channel unbalance problem in feature fusion. As shown in Table II, the detector with SCM achieves a salient improvement in terms of nearly all the evaluation metrics, especially the AP improved from 0.8751 to 0.8843. To better show the effect of SCM, the final feature map F_4 is visualized in Fig. 6. It is obvious that due to the channel unbalance problem in fusion, different-level features cannot be combined well, and many useless channels are included, which causes that the original feature maps contain a lot of noises. By introducing the SCM, the figures in the last column of Fig. 6 show that the interferences are discarded. It indicates that SCM can greatly suppress useless channels and integrate different-level information better. It should be noted that the loss weights with a large difference are set between the two terms because the regression task is more difficult than the classification task. Simultaneously, since the feature area around the boundary is ordinarily beneficial to the regression task, the activation areas almost focus on ship contour.

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TABLE II
ABLATION STUDY OF GRS-DET. ALL MODELS ARE EVALUATED ON THE HRSC2016 VALIDATION SET

Model	Backbone	SCM	Gaussian-Mask	Recall	Precision	F1-Measure	AP
GRS-Det	ResNet-50	×	×	0.9071	0.7997	0.8500	0.8751
		✓	×	0.9169	0.8019	0.8556	0.8843
		×	✓	**0.9413**	0.7511	0.8355	0.8838
		✓	✓	0.9299	0.8311	0.8777	0.8890
	ResNet-101	✓	✓	0.9405	**0.8769**	**0.9076**	**0.8957**

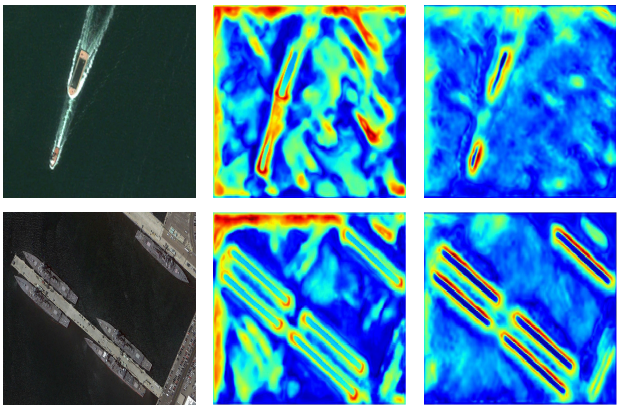

Fig. 6. Visualization of the feature map. (a) Original image. (b) Feature map of GRS-Det without SCM. (c) Feature map of GRS-Det.

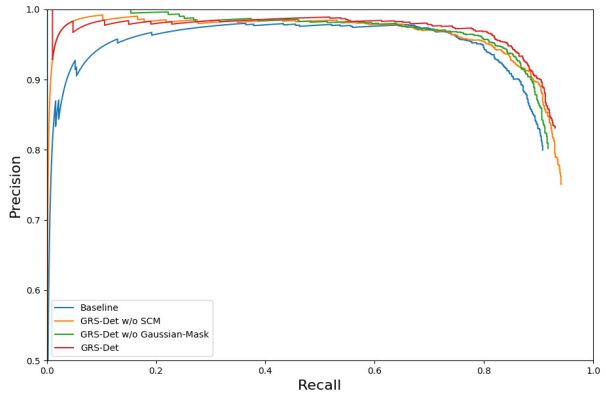

Fig. 7. Precision–recall curve of GRS-Det on the HRSC2016 validation set.

For Gaussian-Mask, we observe a 0.99% increase in AP in the third row of Table II. This indicates that the discriminative capability of the detector is improved by solving the mislabeling problem, which is crucial for obtaining better performance. Furthermore, by utilizing context information, a 3.77% improvement on recall is obtained, which confirms that the detector with Gaussian-Mask can capture more ships.

By adding both SCM and Gaussian-Mask, the experimental results observe a 1.59% improvement on AP. Moreover, it improves the precision while maintaining a high recall, as shown in Fig. 7. This confirms that the detector can capture more ships with the help of Gaussian-Mask, while SCM filters the redundant information and further enhances performance.

TABLE III
OVERALL PERFORMANCE COMPARISONS ON THE HRSC2016 VALIDATION SET

Model	Backbone	Image Size	AP
Fast-RCNN+SRBBS [24]	VGG-16	-	0.5570
BL2 [24]	VGG-16	-	0.6960
R^2CNN [26]	ResNet-101	800×800	0.7307
IENet [43]	ResNet-101	1024×1024	0.7501
RC1&RC2 [24]	VGG-16	-	0.7570
RRPN [25]	ResNet-101	800×800	0.7905
RRD [27]	VGG-16	384×384	0.8430
RoI-Transformer [31]	ResNet-101	512×800	0.8620
Gliding Vertex [33]	ResNet-101	-	0.8820
R^3Det [32]	ResNet-101	800×800	0.8926
GRS-Det	ResNet-50	800×800	0.8890
	ResNet-101	800×800	**0.8957**

To further show the effect of the two modules, the detection results are visualized in Fig. 8. It can be seen that using only SCM can eliminate some background interference denoted by the blue rectangles but lack boundary details denoted by the red rectangles. On the other hand, using only Gaussian-Mask can enhance boundary perception but introduce some noises. Integrating the two modules, it enhances the robustness of the detector and addresses excellent performance even in the complex scene, as shown in the last column of Fig. 8. Finally, we adopt a more powerful backbone (i.e., ResNet-101) in our GRS-Det and achieve the state-of-the-art performance in Table III.

C. Results on HRSC2016

To demonstrate the performance of our method, we compare it with three state-of-the-art methods, i.e., RoI-Transformer [31], R^3Det [32], and Gliding Vertex [33]. Meanwhile, common rotation detectors [24]–[27], [31]–[33], [43] are also used to compare.

The overall comparison performance is reported in Table III. It can be noticed that GRS-Det outperforms the comparative approaches in terms of AP. In comparison with the official baseline of HRSC2016 (BL2 [24]), the performance gain is 28.69%, which denotes the significant superiority of GRS-Det. Compared with the state-of-the-art methods [31]–[33], GRS-Det outperforms them by 3.91%,

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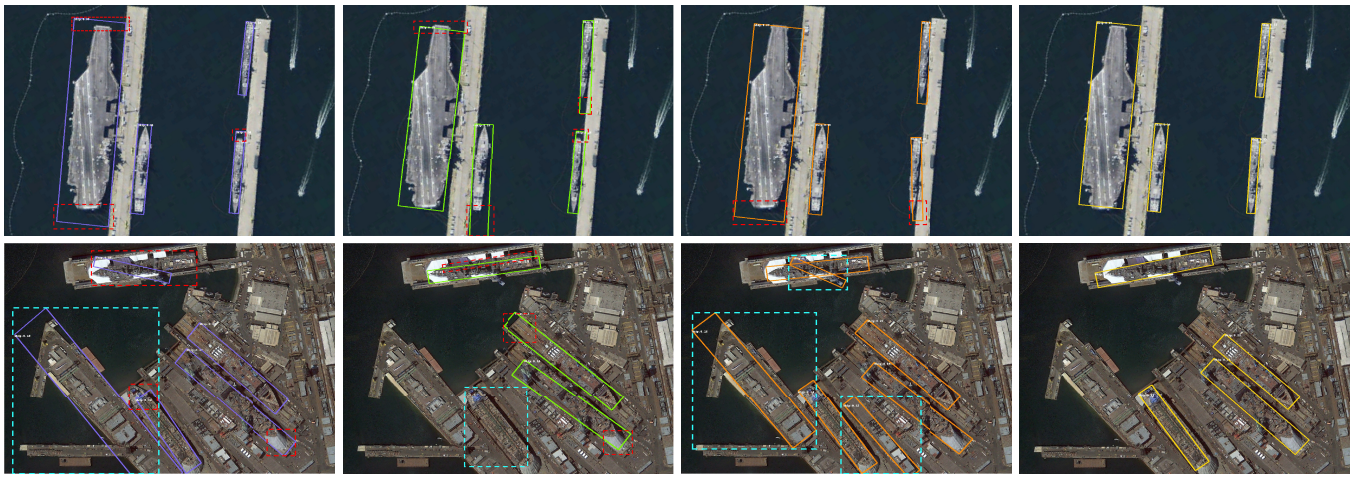

Fig. 8. Detection examples on HRSC2016. We can see that using both SCM and Gaussian-Mask eliminates the background interference denoted by the blue rectangles and enhances the boundary perception denoted by the red rectangles.

1.55%, and 0.35%. This indicates that our method achieves state-of-the-art performance. It is worth noting that GRS-Det under ResNet-50 outperforms almost all the other methods based on ResNet-101. Moreover, the average test time is recorded in Table IV. Our method achieves the best performance on AP with fewer parameters and less test time. In comparison with the anchor-free method [43], under the same test time, our method outperforms it by 18.52% on AP.

To further show the performance of GRS-Det, the detection results are visualized in Fig. 9. As shown in the first row, the ships with a large aspect ratio can be detected well, and the detection boxes are nearly perfect. This results from the context information provided by Gaussian-Mask, which enhances the peripheral details of ships. In the last row, there are a lot of background interferences around ships, which is a huge challenge for ship identification. The results indicate that our proposed method can avoid false detections in the complex environment, which demonstrates the robustness of the detector.

D. Results on DOTA Ship

We also evaluate GRS-Det on the DOTA Ship data set and compare our method with other rotation detection methods [17], [25], [26], [30], [39]. All the comparison experiments are implemented by us.

As can be seen in Table V, GRS-Det achieves 0.8477 AP, which pulls away from the other compared methods. In comparison with the general rotation detection methods [25], [26], [39], our proposed method achieves a remarkable improvement by 0.4023, 0.3253, and 0.1948 in AP. This confirms that the general rotation detectors are not good at processing small and densely distributed ships, while our method achieves a large superiority. Compared with the state-of-the-art methods [17], [30], GRS-Det outperforms them with a 13.28% and 6.29% improvement in terms of AP. This improvement validates the advantages of the proposed method. Moreover, it is shown a 2.16% improvement attained by inserting Gaussian-Mask.

Fig. 10 shows a visual comparison of the detection results of RRPN [25], SCRDet [30], and our method. We find

TABLE IV
PARAMS AND SPEED COMPARISON ON THE HRSC2016 VALIDATION SET

Model	Anchor free	AP	Params	Speed
IENet [43]	✓	0.7501	212MB	**0.0592s**
RRPN [25]	×	0.7905	181MB	0.2857s
RoI-Transformer [31]	×	0.8620	273MB	0.1700s
R^3Det [32]	×	0.8926	227MB	0.0833s
GRS-Det (ResNet50)	✓	0.8890	**127MB**	0.0595s
GRS-Det (ResNet101)	✓	**0.8957**	200MB	0.0729s

that GRS-Det performs well in a variety of challenging scenes. It can be seen that under the densely arranged scene, RRPN [25] loses more overlap objects and produces a lot of false detections. Due to the usage of the attention module, SCRDet [30] can accurately detect ships in the harbor; however, it identifies many cars as ships, as shown in Fig. 10. Compared with the two methods, the GRS-Det demonstrates excellent performance in both dense and sparse scenes. In terms of speed, GRS-Det costs 0.858 and 0.046 s on the dense and sparse scenes in Fig. 9, respectively. The results show that GRS-Det achieves high-precise prediction in dense scenes and high-speed detection in sparse scenes. Due to the pixel-level NMS, the detection time of the GRS-Det is increased in dense scenes, but it still performs best in the average running time on the DOTA ship data set (0.161 s/image).

E. Parameter Analysis

In this part, we conduct two experiments to analyze the effect of shrink rate and loss function. The remaining variable parameters are taken the optimal values, and ResNet-50 is selected as the backbone in all the experiments. The experiments are implemented on HRSC2016 [37].

1) Evaluation of Shrink Rate: The kernel map is the key to recognize the ship object, so how to choose the shrink rate in the label generation has a great effect on performance. In this experiment, the shrink rate is set to 0.2, 0.3, and

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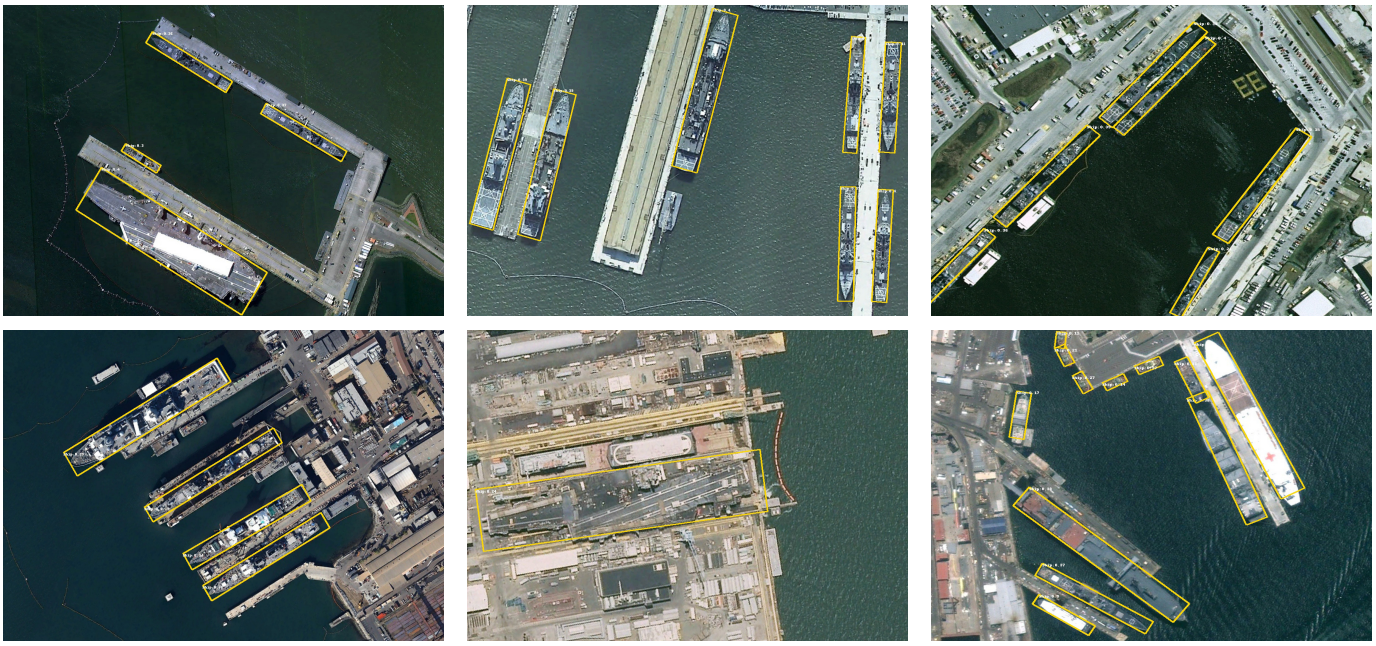

Fig. 9. Visualization of detection results from GRS-Det in HRSC2016. We select the predicted bounding boxes with scores above 0.1.

TABLE V
OVERALL PERFORMANCE COMPARISONS ON THE DOTA SHIP VALIDATION SET

Model	Backbone	Recall	Precision	F1-Measure	AP
RRPN [25]	ResNet-50	0.6527	0.5365	0.5889	0.4454
R^2CNN [26]		0.6834	0.5767	0.6255	0.5224
RetinaNet OBB [39]		0.7856	0.5161	0.6229	0.6529
SCRDet [30]		0.8466	0.6674	0.7463	0.7483
Mask R-CNN [17]		0.8834	0.7448	0.8082	0.7975
GRS-Det w/o Gaussian-Mask	ResNet-50	0.9033	0.8016	0.8494	0.8298
GRS-Det	ResNet-50	**0.9255**	0.8175	0.8681	0.8415
	ResNet-101	0.9130	**0.8401**	**0.8750**	**0.8477**

TABLE VI
PARAMETER ANALYSIS ON SHRINK RATE. ALL MODELS ARE EVALUATED ON HRSC2016

Shrink Rate	Recall	Precision	F1-Measure	AP
0.2	0.9296	0.7282	0.8166	0.8722
0.3	**0.9299**	**0.8311**	**0.8777**	**0.8890**
0.4	0.9055	0.8001	0.8495	0.8846

TABLE VII
PARAMETER ANALYSIS ON THE LOSS RATIO OF KERNEL MAP AND REGION MAP. ALL MODELS ARE EVALUATED ON HRSC2016

λ_k	λ_r	Recall	Precision	F1-Measure	AP
1	0.1	0.9340	0.7877	0.8546	0.8862
1	0.3	0.9381	0.8078	0.8681	0.8883
1	0.5	0.9299	**0.8311**	**0.8777**	**0.8890**
1	0.7	0.9324	0.8195	0.8723	0.8843
1	1	0.9128	0.7790	0.8406	0.8806
0.5	1	**0.9405**	0.7762	0.8505	0.8811

0.4 for comparison. Table VI shows the performance with each shrink rate. It can be seen that when the shrink rate takes 0.2, the interference of the mislabeled edge information will lead to the decline of the precision, and the precision is only 0.7282. When the shrink rate is 0.4, it fails to detect ships with large aspect ratios, which results in a lower recall. When the shrink rate equals 0.3, it correctly detects more ships and gets the highest F1-measure and AP.

To better show the effect of the shrink rate, the kernel map and detection results using different shrink rates are visualized. Through the calculation of the CNN, the kernel map generated during the inference phase is 1/4 of the input image. As shown in Fig. 11, the white area is considered as the place that exists ship. It is obvious that the kernel map achieves the best prediction results when the shrink rate is set to 0.3. With the increase in the shrink rate, since it is hard for the

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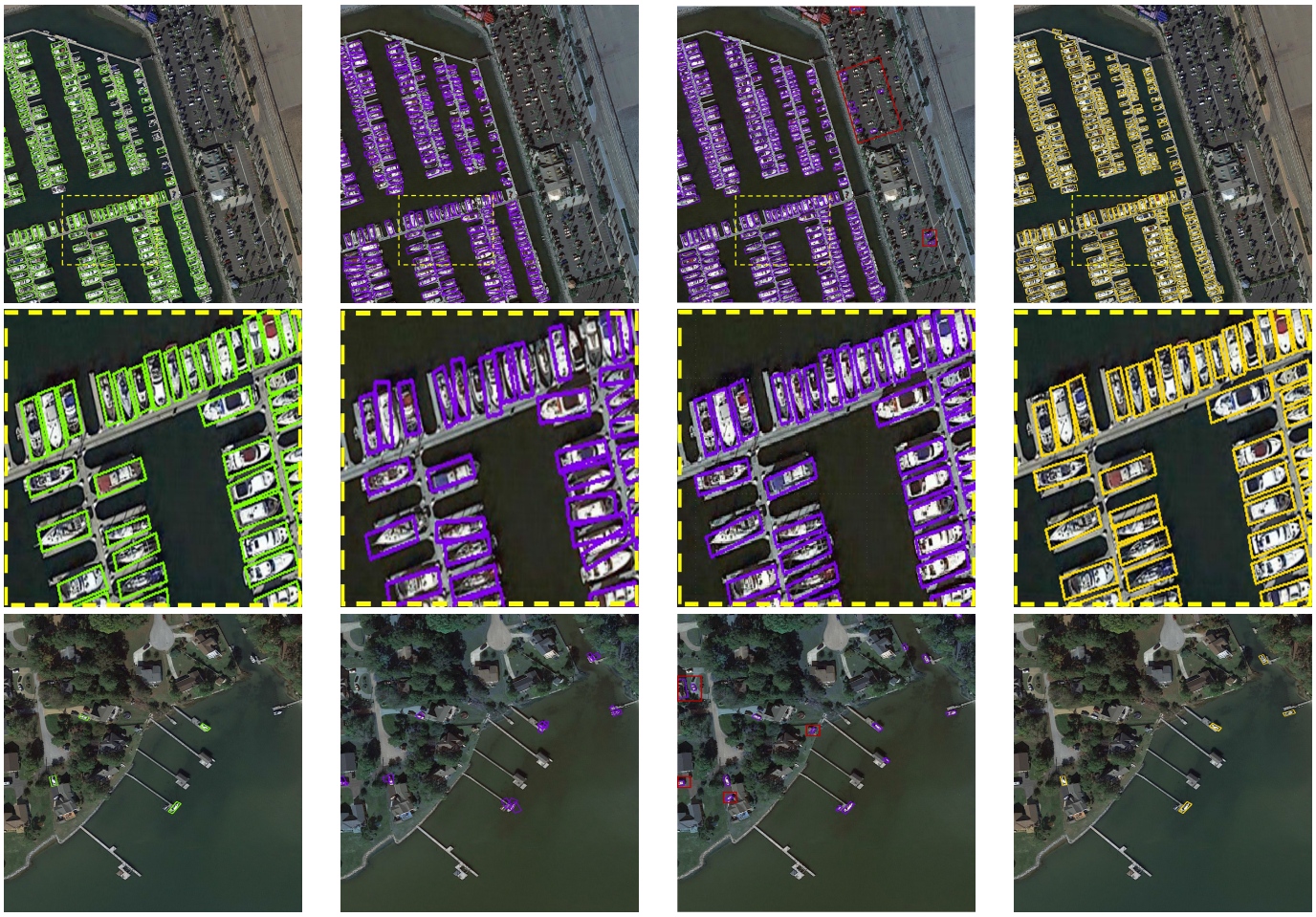

Fig. 10. Comparison results on the DOTA Ship data set. We can see that GRS-Det can better avoid the false prediction results that are indicated by red rectangles. The running times of RRPN [25], SCRDet [30], and GRS-Det are 0.432/0.352, 0.438/0.288, and 0.858/0.046 s, respectively (on the first row/last row images). The average running times of [25], [30], and GRS-Det on the DOTA ship data set are 0.365, 0.270, and 0.161 s, respectively.

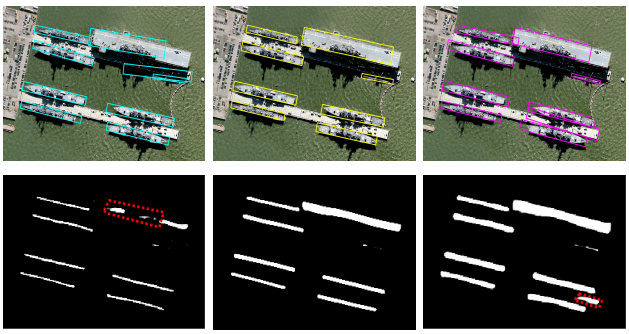

Fig. 11. Visualization of the detection results and kernel map under different shrink rates. (a) Shrink Rate 0.4. (b) Shrink Rate 0.3. (c) Shrink Rate 0.2.

detector to learn enough ship information during the training process, large ships cannot be detected completely in the inference phase, as shown in Fig. 11(a). On the contrary, many mislabeling edge information is introduced with a small shrink rate, which results in more false detections [i.e., in Fig. 11(c)].

2) Evaluation of Loss Function: The loss weights and loss function of the kernel map and region map are essential to detection results. To choose a better combination scheme, two experiments are carried out.

As described in Section III-D, two parameters λ_k and λ_r are used to control the ratio between the kernel map loss and region map loss. In the first experiment, we explore the performance of the different ratio settings. As shown in Table VII, the ratio of 1 to 0.5 exhibits obvious better in both AP and F1-measure. Since the kernel map determines which pixels belonging to ships and the region map only provides context information, it is essential to ensure the kernel map as the dominant. The ratio 0.5 to 1 obtains lower overall performance but performs better in the recall. Although more context information is learned by the region map, it lacks precise discriminate information studied by the kernel map. For the ratio of 1 to 1, neither of the losses dominates; thus, it gets an overall lower result. Therefore, we conclude that it is vital to ensure the kernel map as the dominant, and the region map plays an assist role.

In the second experiment, we evaluate the influence of different loss functions. Table VIII demonstrates the performance of different loss functions. Using the binary cross entropy as the kernel map's loss function, high recall and the lowest

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TABLE VIII
PARAMETER ANALYSIS ON DIFFERENT LOSS FUNCTIONS. ALL MODELS ARE EVALUATED ON HRSC2016

Kernel map	Region map	Recall	Precision	F1-Measure	AP
Dice Loss [48]	Smooth L1	0.9299	**0.8311**	**0.8777**	**0.8890**
	Binary Cross Entropy	0.9234	0.7741	0.8422	0.8821
	Mean Square Error	0.9210	0.7981	0.8552	0.8808
Binary Cross Entropy	Smooth L1	0.9397	0.4852	0.6399	0.8651
Focal loss [39]		**0.9544**	0.6475	0.7715	0.8784

precision are obtained, which indicates that the binary cross entropy is of poor classification ability in this task due to the foreground and background unbalance problem. Although the focal loss [39] proposed to solve class imbalance problem can mitigate the abovementioned problem, the extreme class imbalance in RSIs results in low precision. Meanwhile, GRS-Det is a pixel-based detector whose classification task is similar to the semantic segmentation, and the dice loss [48] can better guarantee the integrity of ships. Therefore, the dice loss [48] is chosen as the loss function in the kernel map to relieve the abovementioned problem. On the other hand, using SmoothL1 in the region map yields much better results than the binary cross-entropy and the mean square error with about 0.78% and 0.93% improvement in AP, respectively. Thus, SmoothL1 is adopted as the loss function in the region map.

VI. CONCLUSION

In this article, we proposed an anchor-free ship detector called GRS-Det. Taking into account the dense distribution of ships, GRS-Det transforms ship detection into a binary semantic segmentation task based on the FCN structure. Meanwhile, an SCM is introduced to feature fusion, which alleviates the channel unbalance problem. For the mislabeling problem in rotation bounding boxes, a rotation Gaussian-Mask is designed to model ship based on its geometry characteristics. Benefited from it, more context information is utilized to strengthen the detection performance. Experiments show that our method achieves state-of-the-art performance on the challenging RS ship detection data sets HRSC2016 and DOTA Ship.

In the future, we will extend our GRS-Det to arbitrary objects to fully explore its potential. In particular, we will construct the Gaussian-Mask for arbitrary objects in RSIs to solve the mislabeling issue, which will further boost the practical value.

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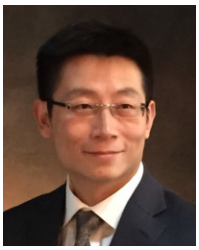

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